A Reduced-Scale Vision Transformer Implementation and CNN Baseline Evaluation on CIFAR-10

Aditya Kumar Jha
Independent Course Project
[Add institutional affiliation/email if required]

Abstract

Vision Transformers (ViTs) apply self-attention to sequences of image patches and provide an alternative to convolutional neural networks for visual recognition. This project implements a compact ViT from scratch in PyTorch and evaluates it on CIFAR-10. A CIFAR-adapted ResNet-18 is trained as a convolutional baseline. Additional experiments investigate reduced training-data availability and the effect of removing dropout. The ViT contains 1,205,898 trainable parameters and reaches 72.00% test accuracy after 15 epochs, while ResNet-18 contains 11,173,962 parameters and reaches 89.41%. Training the ViT on 20% of the training partition (9,000 images) yields 51.11% test accuracy. A single run with dropout set to zero reaches 75.02%, compared with 72.00% for the baseline with dropout 0.1. These results indicate that, under this compact from-scratch training setup, the convolutional baseline is more accurate and the ViT is sensitive to training-set size. The dropout finding is preliminary because the comparison is based on single runs with different random seeds. This study is a reduced-scale educational reproduction and does not attempt to reproduce the original ViT paper's large-scale training regime.

Keywords— Vision Transformer, ViT, CIFAR-10, self-attention, ResNet-18, ablation study, image classification.

I. INTRODUCTION

Convolutional neural networks (CNNs) use local connectivity and weight sharing to encode useful spatial inductive biases. Vision Transformers adapt the Transformer architecture to images by dividing an image into patches, projecting each patch into an embedding, and processing the resulting token sequence with self-attention [1]. The original ViT study demonstrated that Transformers can achieve strong image-recognition performance when trained with sufficient data and appropriate procedures.

This project implements the central ViT components directly in PyTorch and studies their behavior on CIFAR-10, a ten-class image-classification benchmark [2]. Because CIFAR-10 images are only 32 × 32 pixels and compute is limited, the model and training budget are deliberately smaller than those in the original work. The experiments ask: (1) how does a compact ViT compare with a CNN baseline; (2) how sensitive is ViT performance to reducing the training set; and (3) what is the observed effect of removing dropout?

II. METHOD

A. Dataset and preprocessing

CIFAR-10 contains 60,000 color images of size 32 × 32 distributed across ten classes. The 50,000-image training partition was split into 45,000 training images and 5,000 validation images using a fixed seed. The official 10,000-image test partition was kept separate. Training preprocessing used random cropping with four-pixel padding, random horizontal flipping, tensor conversion, and channel normalization. Validation and test images were normalized without random augmentation.

For the low-data experiment, 20% of the 45,000-image training partition was sampled, resulting in 9,000 training images. Validation and test partitions remained unchanged.

B. Vision Transformer architecture

Each image is divided into non-overlapping 4 × 4 patches. The number of patches is $N = (32/4)^2 = 64$. A learned classification token is prepended, giving 65 tokens. A convolution with kernel size and stride equal to the patch size implements patch projection. Learned positional embeddings are added to preserve patch-order information.

The model uses embedding dimension D = 128, four attention heads, six Transformer encoder blocks, MLP hidden dimension 512, and baseline dropout 0.1. Each block uses pre-layer normalization, multi-head self-attention, a residual connection, and a feed-forward MLP with GELU activation. The final normalized classification-token representation is passed to a linear classifier with ten logits. The model has 1,205,898 trainable parameters.

Scaled dot-product attention is Attention(Q,K,V) = softmax(QK■/√d■)V, where Q, K, and V are query, key, and value projections and d■ is the per-head key dimension. Multiple attention heads are concatenated and projected back to the embedding dimension.

C. CNN baseline

ResNet-18 [3] is initialized from scratch. To accommodate CIFAR-10 resolution, its initial convolution is replaced with a 3 × 3 convolution of stride one, and initial max-pooling is removed. The final fully connected layer outputs ten classes. The adapted model has 11,173,962 trainable parameters.

D. Optimization and checkpointing

Primary experiments used 15 epochs, batch size 64, AdamW with learning rate 3 × 10■■ and weight decay 0.05, and cosine annealing with a 15-epoch schedule. The checkpoint with the highest validation accuracy was selected, and final metrics were evaluated on the held-out test set. Training ran locally on Apple Silicon using PyTorch MPS. Checkpoints and per-epoch history were saved after every completed epoch.

III. EXPERIMENTS

Four experiments were completed: (1) full-data ViT with dropout 0.1; (2) full-data ResNet-18 trained from scratch; (3) ViT trained using 9,000 images; and (4) full-data ViT with dropout set to zero. The dropout ablation is exploratory: the zero-dropout and baseline runs used different random seeds and were not repeated over multiple seeds.

IV. RESULTS

Table I reports the saved local-run results. The best epoch is selected using validation accuracy; test accuracy is reported for the selected checkpoint.

TABLE I. CIFAR-10 RESULTS

Experiment	Params (M)	Best epoch	Val. (%)	Test (%)
ViT, dropout 0.1	1.206	15	71.94	72.00
ResNet-18, scratch	11.174	14	90.34	89.41
ViT, 20% data	1.206	15	51.22	51.11
ViT, dropout 0.0	1.206	15	74.42	75.02

A. ViT versus ResNet-18

ResNet-18 achieved 89.41% test accuracy, compared with 72.00% for the ViT, an absolute difference of 17.41 percentage points. ResNet-18 has approximately 9.27 times as many trainable parameters. This result is consistent with the possibility that convolutional inductive biases are advantageous for a small image dataset and limited training budget. However, the experiment compares only these implementations and hyperparameters; it does not establish a general ranking between CNNs and Transformers. ResNet-18 reached its best validation accuracy at epoch 14, while the baseline ViT was best at epoch 15, so the fixed budget may not represent convergence.

B. Effect of reduced training data

Reducing the training partition from 45,000 to 9,000 images decreased ViT test accuracy from 72.00% to 51.11%, a decline of 20.89 percentage points. This indicates substantial sensitivity to training-set size in this from-scratch setting. The result should be interpreted under this subset selection, augmentation pipeline, and training budget.

C. Dropout ablation

The zero-dropout run reached 75.02% test accuracy, 3.02 percentage points higher than the baseline's 72.00%. Validation accuracy increased from 71.94% to 74.42%. This is evidence that removing dropout was beneficial in this individual run, but it is not sufficient to establish that dropout is generally harmful. The runs were not paired across identical initializations and have not been repeated over multiple seeds. A controlled multi-seed ablation should report mean and standard deviation.

V. ATTENTION ROLLOUT

Attention rollout was implemented by averaging attention across heads, adding identity connections to represent residual paths, row-normalizing matrices, and composing attention matrices across layers. Scores from the classification token to patch tokens were reshaped into an 8 × 8 grid and upsampled for visualization. In the inspected example, the true and predicted class were both cat. This verifies that the visualization pipeline executed, but a single example does not establish explanation faithfulness or overall predictive performance. Attention rollout is a qualitative diagnostic rather than a causal explanation.

VI. LIMITATIONS AND FUTURE WORK

This is a reduced-scale experiment on CIFAR-10 rather than the large-scale ImageNet/JFT setting used in the original ViT work. The primary comparison uses one run per configuration and only 15 epochs. The no-dropout result is based on a single run and should be repeated with matched seeds and multiple independent runs. The study does not include a pretrained-backbone experiment. The baseline ViT's best validation epoch was the last epoch, suggesting that additional training may improve performance. Future work should extend the training budget for both architectures, perform a multi-seed dropout ablation, report mean and standard deviation, and optionally compare a pretrained model as a separate transfer-learning setting.

VII. CONCLUSION

A compact ViT was implemented from scratch and evaluated on CIFAR-10 with a ResNet-18 baseline, reduced-data experiment, dropout ablation, and attention rollout. The ViT achieved 72.00% test accuracy, while adapted ResNet-18 achieved 89.41% under the same 15-epoch budget. Training the ViT with 20% of the training partition reduced test accuracy to 51.11%. A single no-dropout run reached 75.02%, motivating but not proving a potential benefit from reducing dropout. Overall, the project demonstrates the ViT pipeline and provides an empirical comparison while highlighting the importance of dataset size, inductive bias, and replication.

REFERENCES

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